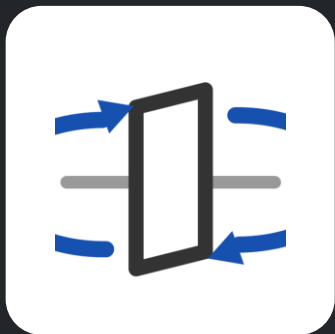




Solve GJ

Legacy: recompute the torsional stiffness GJ of an existing EI and Cross Section feature and write it back to its stored data and table.

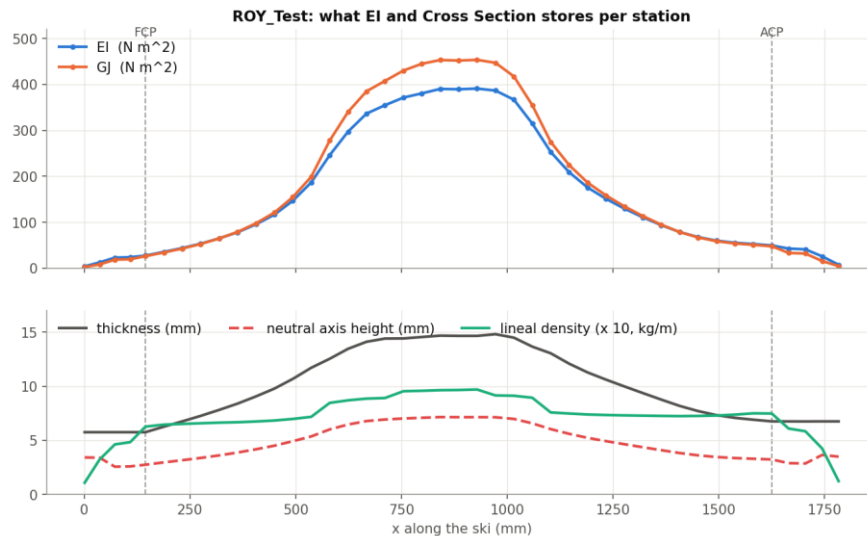


What it does

GJ used to be a separate step. EI and Cross Section now computes GJ itself with the same solver, so Solve GJ changes nothing on a current model. It is kept so older documents still regenerate.

Use it when

- An older document still has it in the tree
- Not needed for new work



GJ (orange) is already in every EI and Cross Section result.



The dialog

1

SG1 Solve GJ on X1 -> O...  

Cross section feature
X1 EI_iso_rect_100x10 E 10 G... x



#	Parameter	What it means
1	Cross section feature	One EI and Cross Section feature.

$GJ = 4 \times \sum (G \times I_z)$ over the section's triangles, G = each body's Q66 (thin-plate Saint-Venant).



Outputs and messages

Outputs

Key	What it holds
(stored data)	GJ of every station in the feature's CrossSectionAnalysis data and table.

Messages

	Message / what to do
Error	Select the cross section feature. Nothing selected.
Error	(missing data / station count mismatch) The selected feature has no stored sections.



Tips and limits



Fixed 2026-09-25

The recomputed GJ was written to a copy and lost; errors were swallowed. Both fixed.



Redundant

Safe to delete from new models.



Where it lives

Document	xSection (f8deedeb1fbd819a8fa20113), 'Stiffness Tools'
Tab	GJ_Feature (+ section/gjAnalysis.fs, gjDataAccess.fs)
Tests	SG1 in the xSection tests studio (stored GJ 125.31)
Examples	xSection tests: SG1
Explainer	docs/explainers/xsection/xsection_explained.md, section 1.5, 2.2
Related	EI and Cross Section, Estimate Stiffness, Estimate Deflection, Generate / Analyze baseline, Update profile